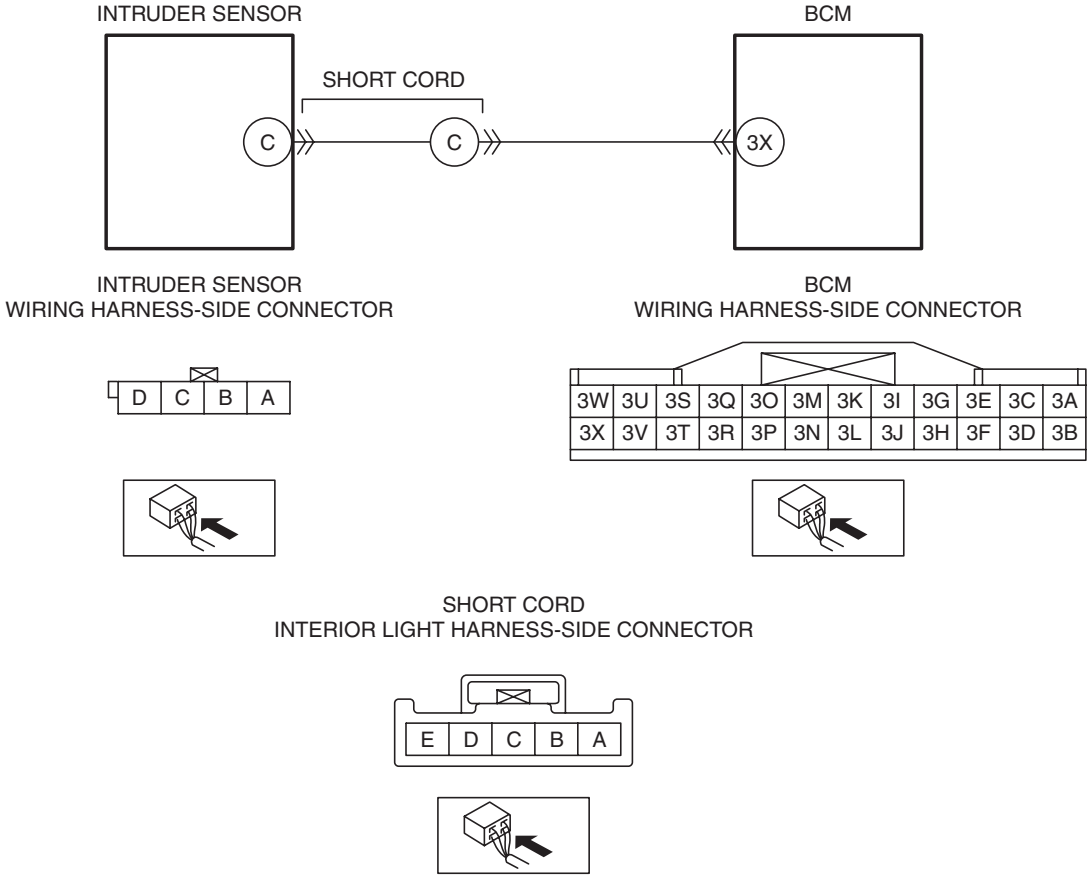
DTC B109F:87 [BCM]

id0902f5412800

Description	Communication error with intruder sensor
Detection condition	• BCM cannot receive the signal from the intruder sensor for **1.5 s or more** with the ignition switched ON (engine off or on).
Fail-safe	• Not applicable
Possible cause	• Short cord connector or terminal malfunction • BCM connector or terminal malfunction • Short to ground in wiring harness between short cord terminal C and BCM terminal 3X • Open circuit in wiring harness between short cord terminal C and BCM terminal 3X • Intruder sensor malfunction • BCM malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY BCM DTCs AGAIN** • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Lock the door to go on theft-deterrent system standby. • Unlock the door to cancel theft-deterrent system standby. • Switch the ignition ON (engine off or on) and wait for **1.5 s or more**. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Go to the next step.
		No	Go to Step 8.

Step	Inspection		Action
2	**INSPECTION SHORT CORD CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the short cord connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
3	**INSPECT BCM CONNECTOR CONDITION** • Disconnect the BCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
4	**INSPECT INTRUDER SENSOR CIRCUIT FOR SHORT TO GROUND** • Verify that the short cord and BCM connectors are disconnected. • Inspect for continuity between short cord terminal C (interior light harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between short cord terminal C and BCM terminal 3X. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 7.
		No	Go to the next step.
5	**INSPECT INTRUDER SENSOR CIRCUIT FOR OPEN CIRCUIT** • Verify that the short cord and BCM connectors are disconnected. • Inspect for continuity between short cord terminal C (interior light harness-side) and BCM terminal 3X (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between short cord terminal C and BCM terminal 3X. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 7.
6	**INSPECT INTRUDER SENSOR** • Inspect the intruder sensor. (See INTRUDER SENSOR INSPECTION.) • Is the intruder sensor normal?	Yes	Go to the next step.
		No	Replace the intruder sensor, then go to the next step. (See INTRUDER SENSOR REMOVAL/INSTALLATION.)

Step	Inspection		Action
7	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Lock the door to go on theft-deterrent system standby. • Unlock the door to cancel theft-deterrent system standby. • Switch the ignition ON (engine off or on) and wait for **1.5 s or more**. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
8	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [BCM].)
		No	DTC troubleshooting completed.